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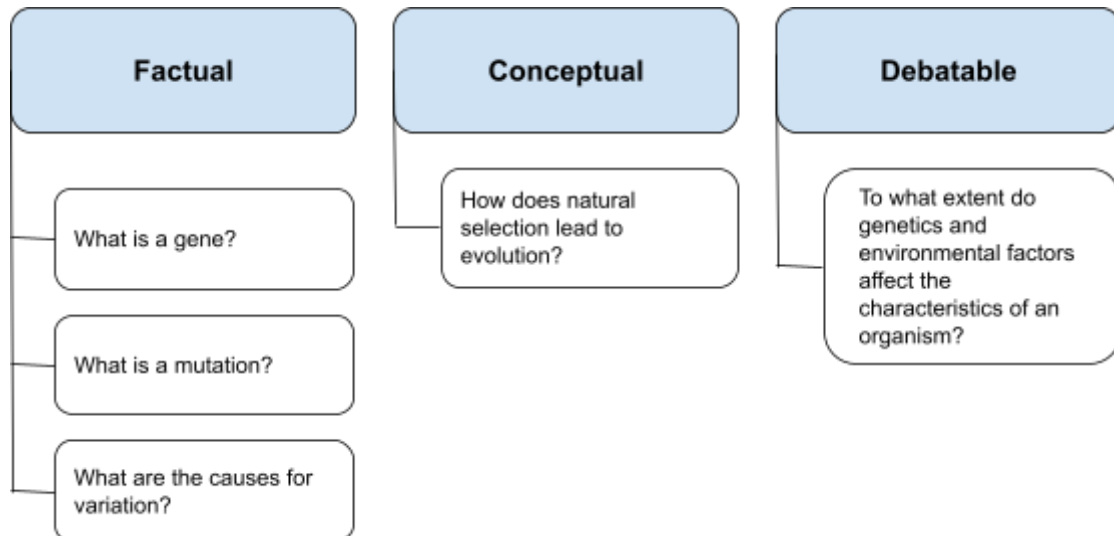


Inheritance & Variation

Lesson No.	Content	Lesson No.	Content
1	Variation	6	Genes and Mutation
2	Selective Breeding	7	Genetic Crosses
3	Natural Selection	8	Genetic Disorders
4	Introduction to DNA	9	Genetic Engineers
5	DNA Extraction	10	Criterion A Assessment

Statement of Inquiry: Evolution is a consequence of change in the environment that affects inheritance.

In this unit, we will consider these questions:



These concepts will be covered...

Key concept: Change

Related concepts: Consequences, Environment

Global Concept: Scientific and technological innovation

This unit will give the opportunity to improve in these skills...

- Critical-thinking skills
- Information Literacy

Assessment opportunities in this unit include...

Criterion A: Knowing and Understanding

Keywords list

Inherited characteristics/How variation occurs

Word	Pronunciation	Meaning
characteristics	Features of an organism, e.g. eye colour, height or blood group.	
DNA	Deoxyribonucleic acid - the molecule that carries the genetic information in the nucleus of every cell.	
gamete	A sex cell (sperm cell or egg cell) that carries half of genetic information.	
gene	A section of DNA that codes for a particular characteristic (a protein).	
genetic information	The instructions stored in DNA that control how an organism develops and its characteristics.	
inherited	Passed on from parents to their offspring through genes.	
variation	The differences between individuals of the same species.	
Mutation	A random change in the DNA (a gene) that may alter a characteristic.	

Different varieties/Environmental effect

Word	Pronunciation	Meaning
environment		
environmental factors		Caused by the environment, e.g. scars, an accent, or a plant growing tall in good light.
resistant		Not harmed or killed by something, e.g. a bacterium that an antibiotic cannot kill.
species	spee-shees	A group of organisms that can breed together to produce fertile offspring.
variety		A group of organisms within a species that share some different characteristics.

Animal breeding

Word	Pronunciation	Meaning
breed		A set of animals that are in some way different from other members of the same species.
breeding		To mate two organisms of the same species to produce offspring.
cross-breeding		When different varieties or breeds are mated with one another.
selective breeding		When humans choose certain animals and plants that have useful characteristics and breed more of these organisms.
yield		How much of something useful to humans that an organism produces.

Inheritance

Word	Pronunciation	Meaning
allele		Different versions of the same gene, e.g. alleles for blue or brown eye colour.
homozygous		Having two identical alleles for a gene, e.g. BB or bb.
heterozygous		Having two different alleles for a gene, e.g. Bb.
genotype	<i>jeen -o-type</i>	The alleles an organism has for a characteristic, e.g. Bb.
phenotype	<i>feen-o type</i>	The characteristic you can see, e.g. brown eyes, produced by the genotype.
dominant		An allele that is always shown in the phenotype, even when only one copy is present.
recessive		An allele only shown in the phenotype when two copies are present (no dominant allele).
punnett square		A grid used to work out the possible genotypes of offspring from two parents.

Lesson 1: Variation

Learning Objectives:

- Define the term *variation*.
- Discuss the fact that variation occurs within as well as between species.
- Describe the differences between continuous and discontinuous variation, using examples of a range of characteristics found in plants, animals and microorganisms.
- Explain both genetic and environmental causes of variation.

Task

Variation exists in all species. Inheritance is the science of how all species inherit our traits and therefore maintain variation within species.

Gregor Mendel was one of the first scientists to experiment with the nature of Inheritance.



Research what Gregor Mendel did and why he is called 'the father of genetics.'

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Mendel was a monk who bred pea plants and tracked their characteristics. He showed that 'factors' (now called genes) are inherited from both parents in predictable ratios, so he is called the father of genetics.

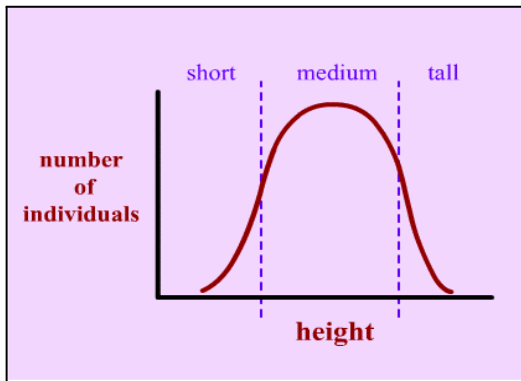
Types of Variation

The traits that Mendal was looking at could be in two categories, the same as any trait in humans too. Define the terms below.

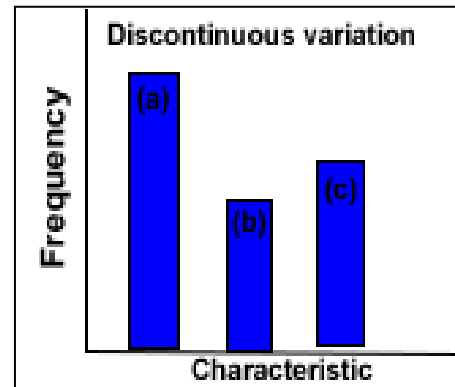
Continuous Variation ~~Variation in a characteristic that can take any value in a range, e.g. height or mass.....~~

Discrete Variation ~~Variation in a characteristic that falls into separate groups or categories, e.g. blood group.....~~

Continuous



Discrete



What type of graph would we use to plot discontinuous data?

On the next page, plot a bar graph of this data.

~~Continuous (e.g. height): values spread over a range, shown as a line graph/histogram. Discrete (e.g. blood group): separate groups, shown as a bar chart.~~

Blood Group	Percentage of population
A	42
B	10
AB	4
O	45

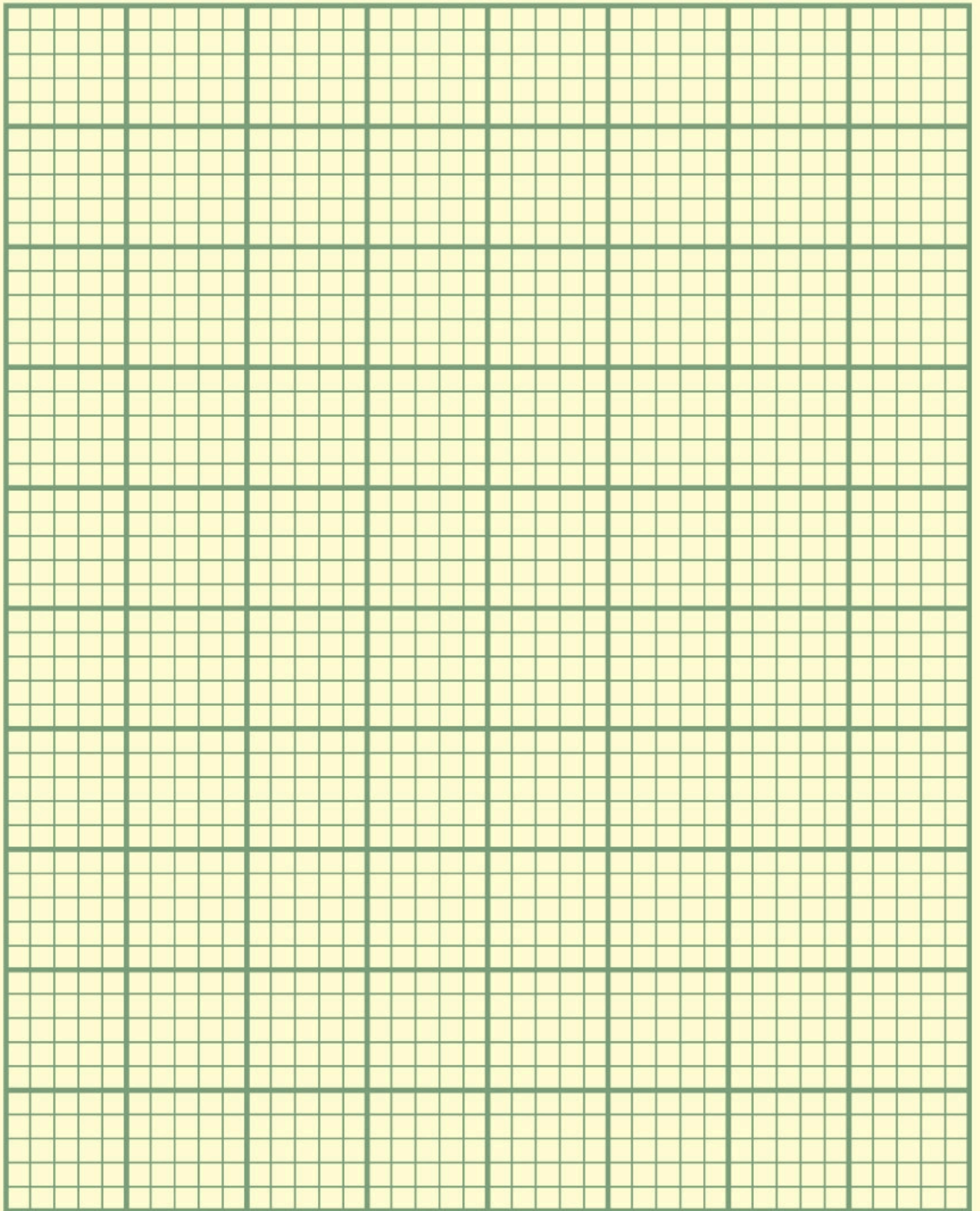
A bar chart, because the data falls into separate groups (categories).

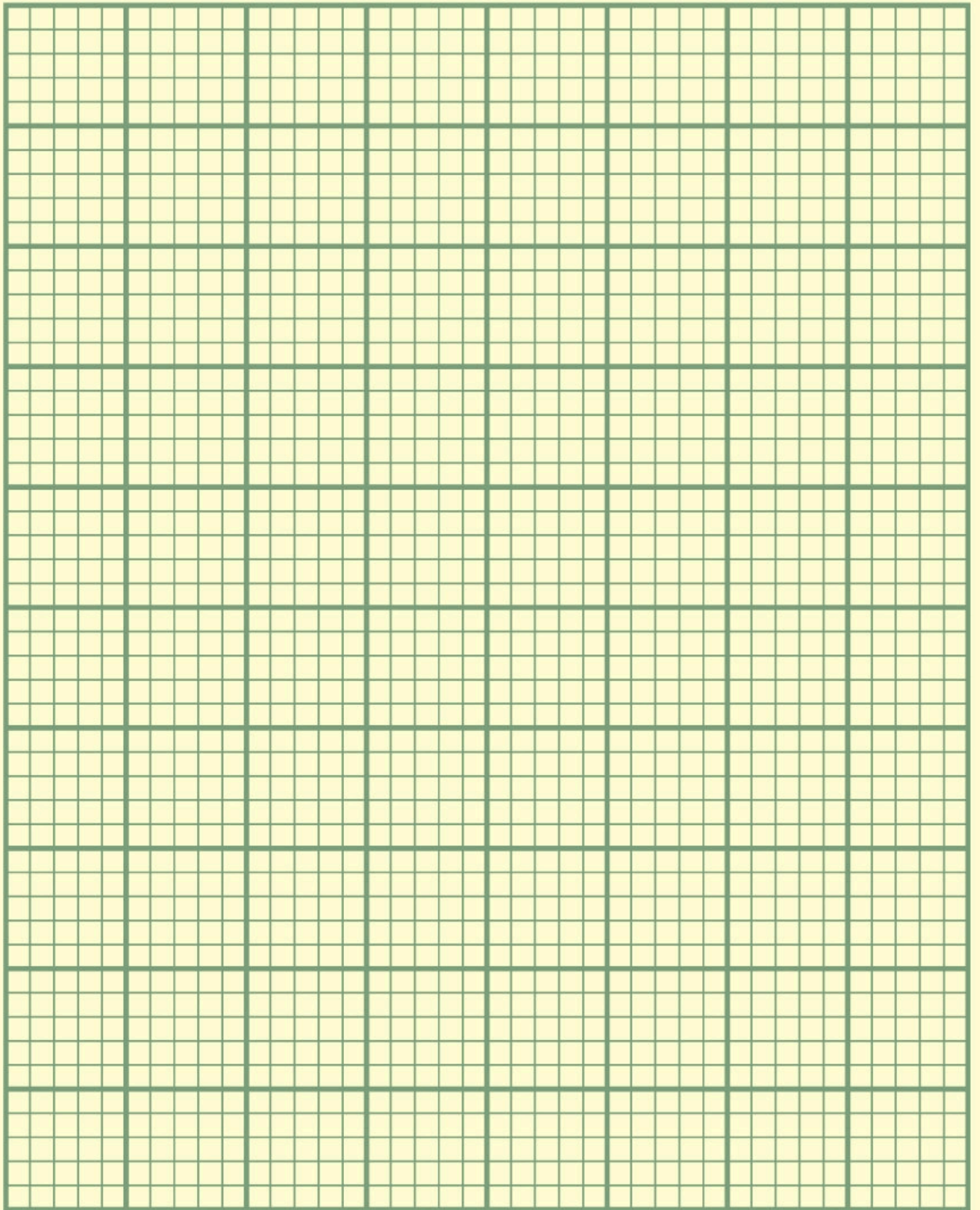
What type of graph would we use to plot continuous data?

On the next page also, plot a line graph of this data.

Height (cm)	Number of people
Up to 129	0
130-134	5
135-139	9
140-144	16
145-149	10
150-154	6
155-159	4
Over 160	1

A line graph (or histogram), because the data can take any value within a range.





Some pupils measured the masses of two different varieties of apple. There were five apples of each variety. The results are recorded in the table.

	Variety 1 mass (g)	Variety 2 mass (g)
Apple 1	125	150
Apple 2	120	147
Apple 3	122	150
Apple 4	130	152
Apple 5	125	158

Variety 2 has different alleles (a different genotype) that make it grow larger, so the difference is inherited, not caused by the environment.



- 1 Draw a bar chart to show the differences in mass for variety 1.
- 2 Why do you think the variety 1 apples have different masses?
- 3 Plot the masses for variety 2 on the same chart.
- 4 Why do you think the variety 2 apples have different masses?
- 5 Why do you think variety 2 apples have larger masses than variety 1 apples?

Bar chart: apple 1-5 on the x-axis, mass (g) on the y-axis - bars of 125, 120, 122, 130, 125 g for variety 1.



knowledge, presenting, considering



Some pupils measured the masses of two different varieties of apple. There were five apples of each variety. The results are recorded in the table below.

	Variety 1 mass (g)	Variety 2 mass (g)
Apple 1	125	150
Apple 2	120	147
Apple 3	122	150
Apple 4	130	152
Apple 5	125	158

The apples have different genotypes (alleles) and grew in slightly different conditions, so they took in different amounts of water and nutrients and grew different masses.



- 1 Enter the data in a spreadsheet.
- 2 Use the spreadsheet to draw a bar chart to show the range in mass for variety 1.
- 3 Use the spreadsheet to draw a bar chart to show the range in mass for variety 2.
- 4 Why do you think apples within a variety have different masses?
- 5 Work out the mean mass of each variety and plot this information on the chart.
- 6 Why do you think variety 2 is different to variety 1?

Optional extra

- 7 Calculate the following averages for these apples and explain what each average shows.

a mean b median c mode

Add bars for variety 2 on the same chart: 150, 147, 150, 152, 158 g (side by side with variety 1).



knowledge, numeracy, presenting, considering



Lesson 2 - Selective Breeding

Learning Objectives

- Define Selective Breeding
- Explain the process, using examples from different species
- Discuss the advantages and disadvantages of the process

✓ True or False

1. The color of your eyes is a characteristic that is inherited.
2. You inherit characteristics from your mother only.
3. Your brothers or sisters inherit exactly the same characteristics as you.
4. All animals inherit some characteristics from their parents.
5. Plants are not able to inherit characteristics.
6. Not all characteristics are inherited.
7. The environment has no effect on your characteristics.
8. Genes carry genetic information and are found inside every cell.
9. There are half as many genes in gametes as there are in normal cells.
10. Fertilisation is when a sperm cell fuses with an egg cell.
11. A fertilised egg cell only contains half the genetic information of a normal body cell.

Selective Breeding

Roughly 10 000 years ago ~~humans started to farm and keep plants and animals.~~

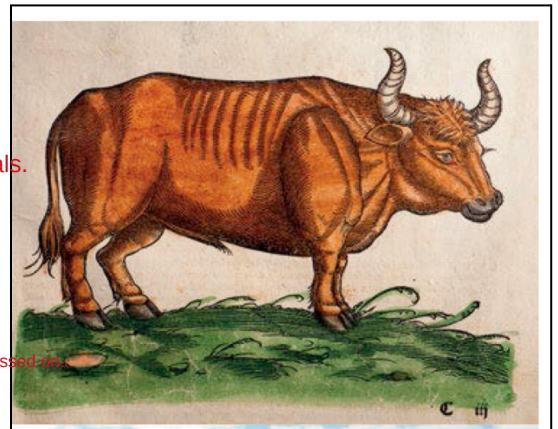
One of the first animals farmed was the Auroch. They were gathered into herds and over hundreds of years the ones that

~~were the most useful (or biggest or fastest) were kept and bred together so their useful characteristics were passed on.~~

Now the auroch is extinct but its descendants, the modern cow, is one of the most farmed animals on the planet.

~~This process is called selective breeding - humans choose which organisms breed~~

It is how we have got the modern day cow



Dog Breeding

Ever wonder how your dog evolved? There are more than 350 different breeds of dog, each one - from the Chihuahua to the Great Dane - descended from the Grey Wolf.

Over the thousands of years, we have shaped wolves through breeding into dogs with floppy ears, sleek coats, and short noses - exactly the opposite of their distant ancestors.

We have created dogs to suit our needs. This process of breeding dogs for different useful characteristics is called **selective breeding**.

You will be given a picture of a dog. Answer these questions about the dog:

1. What useful characteristics does the dog have?

.....
.....
Depends on the dog given - e.g. for a greyhound: long legs, slim body, strong heart and lungs for speed; for a St Bernard: large size and thick coat for cold mountains.

2. Write down the reasons why the dog was bred in the past e.g. hunting, companionship?

3. Give some reasons why the breed is still popular today.

.....
.....
For work in the past: hunting, herding livestock, guarding property, pulling sledges, and for companionship.

Now you will do some selective breeding.

Decide which two dogs you will breed to create a new breed of dog that will suit the jobs below:

1. Dog needed for racing. Must be very fast but medium sized (50cm tall). An enthusiastic worker and also a good hunter.

It is still popular as a pet/companion, for dog shows, for sport (racing, agility), for guarding and for working dogs such as police or rescue dogs.

2. Very intelligent dog that has an excellent sense of smell is needed. Must be a large dog (65cm tall) and live a long life (more than 12 years).

Cross a Greyhound (very fast, keen hunter) with a medium-sized working breed such as a Foxhound - offspring should be about 50 cm, fast, keen and hard-working.

3. Large (60cm tall), strong, muscular dog wanted for large game hunting. Must be able to work in a team and be away from home for long periods of time.

Cross a German Shepherd (intelligent, large, keen nose) with a Labrador Retriever (excellent sense of smell, long-lived) to get a large, clever, sniffing dog.

4. Kind, patient, good-looking, long-haired dog wanted. Preferably light coloured fur and large (65-70cm tall). Must be easy to train, obedient and able to get on with other people including children and strangers.

Cross a Rottweiler (large, strong, muscular, guard instincts) with a Rhodesian Ridgeback (large-game hunter that works in a team away from home).

5. Large dog (65-70cm tall), must like water and be a good swimmer and hunter. Must have good stamina and be an excellent guard-dog.

Cross a Newfoundland (large, loves water, strong swimmer, guard instincts) with a German Shepherd (stamina, excellent guard dog) - offspring like water, swim, hunt and guard.

6. Small (35-40cm tall) very friendly dog required as a companion to children. Long white hair is preferable. Needs to be welcoming to strangers and able to live outside.

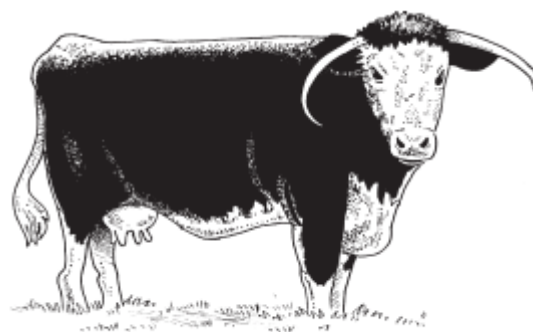
Cross a Bichon Frise or Maltese (small, long white hair, friendly) with a small, friendly breed such as a Cavalier King Charles Spaniel (good with children, welcomes strangers).

7. Dog required that must be an excellent hunter (very fast) and retriever of game. Must be medium sized (around 55cm tall) with a short haired coat and definitely not slobber or smell.

Cross a Whippet/Greyhound (very fast hunter) with a Pointer or Labrador (retriever of game, medium size, short coat, no smell) to get a fast, tidy 55 cm retriever.

Rare Breeds L7-9

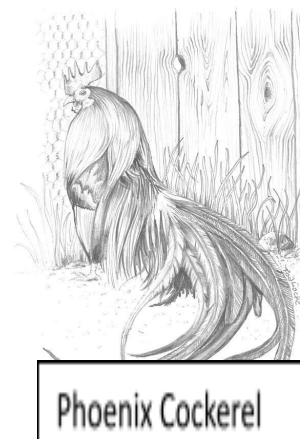
Popular farm animals now have been created from old breeds through selective breeding. The old breeds do not have characteristics that farmers now want. So they do not go extinct they are kept in rare breed farms. A good example are Shire horses (large work horse) they were used to pull ploughs. They are no longer needed as we now have machines and tractors for that.



An English longhorn cow.

Animals at rare breeds farms are involved in breeding programs to ensure the species lives on. It is important they are kept in case their characteristics are needed again.

Robert Bakewell in the 18th century bred English Longhorns. He was the first to selectively breed cattle for farmers. He selected for large rear ends and quick growth. English longhorns were popular until the early 19th century after this the short horn became popular.



Phoenix Cockerel

Each breed of animal has certain characteristics and a breed is only classed as a breed if all of its offspring have these same characteristics. It has to exist for 6 generations or more (roughly 75 years).

Questions:

1. How are new breeds of farm animal created?
2. Where are old breeds kept?
3. Why do we keep rare breeds ?
4. Why are shire horses no longer needed?
5. How long does a breed need to exist before it is called a breed?
6. How would you know if something is a new breed or not?
7. What characteristics were selected in the English longhorn?
8. Why are they useful?
9. Suggest why the shorthorn became more popular
10. Which feature has the phoenix cockerel been selected for?
11. Why do farmers no longer breed animals that just look interesting?

By selective breeding, farmers choose animals that have the useful characteristics and breed them together, repeating this over many generations.

Old breeds are kept on rare-breed farms:

So that they do not go extinct - their characteristics may be needed again in the future.

Because machines and tractors are now used for ploughing instead of horses.

It must exist for 6 generations or more (roughly 75 years).

It is a new breed only if ALL of its offspring have the same characteristics as the parents, and it has existed for 6+ generations.

A large rear end (lots of meat) and quick (fast) growth.

They produce more meat (beef) and grow quickly, so farmers get a higher yield and more income.

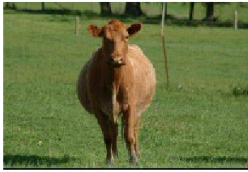
The shorthorn probably produced more meat or milk (better yield) or was easier/cheaper to keep, so farmers preferred it.

The phoenix cockerel has been selected for its very long tail feathers (an ornamental feature).

Farmers must make a profit, so they now breed for characteristics that increase yield (meat, milk, wool) rather than just looking interesting.

SELECTIVE BREEDING IN COWS

Look at the cows below:



Gertrude - produces extra creamy milk, not much meat



Betsy - has low milk production but lots of tasty meat



Penelope - Has little meat but produces milk that contains lots of extra vitamins and minerals



Alma - lots of meat, good milk production but milk sours quickly.

Look at the bulls below:



Bruiser - can be violent. Has lots of muscle. Cannot survive hot, dry conditions but does very well in damp climate



Rufus - Can cope extremely well in hot, dry conditions. Has little meat but it is very tasty.



Donald - has lots of meat, can survive well in temperate climates. Comes from a family line of cattle that produce high quantities of milk.



Boris - has lots of muscle but can cope with hot, dry conditions. Family line does not produce much milk.

Questions

Which bull and cow would the farmer mate together to get:

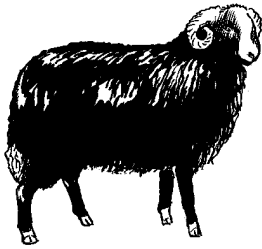
1. Lots of meat and creamy milk. The animal should be able to survive in hot dry climates.
2. Lots of milk with vitamins and minerals and can survive well in hot dry climates
3. Can produce extremely expensive delicious meat. No need to produce milk for drinking
4. Can survive in temperate climate and produces lots of milk and meat

Mate the bull that produces lots of meat with the cow that gives creamy milk, choosing animals that can both survive in hot, dry climates.

Mate the bull whose relatives give lots of milk with vitamins and minerals (or that survives hot dry climates) with the cow that gives lots of milk and survives hot dry climates.

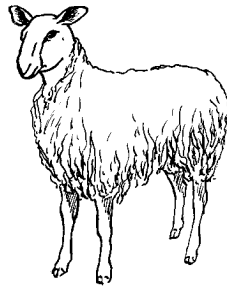
Mate the bull and cow that are both from meat-producing breeds that produce extremely expensive, delicious meat (dairy traits are not needed).

Selective Breeding



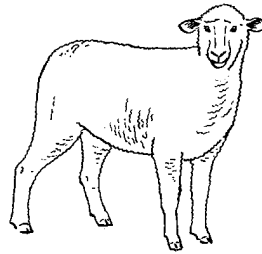
A

The Balwen mountain sheep is able to live in cold, harsh conditions.



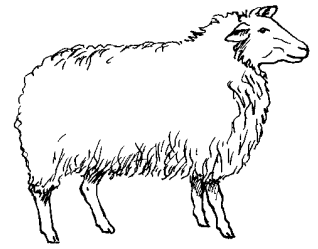
B

The Bluefaced Leicester sheep produces good quality wool.



C

The Polled Dorset sheep produces good quality meat.



D

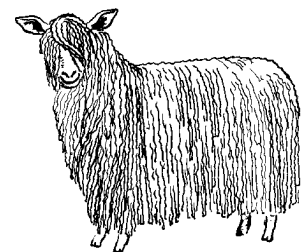
The Friesian Milk sheep produces a lot of milk.

1. Each of these sheep has been produced by **selective breeding**. For each sheep write down one characteristic that has been selected

Farmers often want to breed two different breeds of sheep together. This is called **cross-breeding**. The offspring produced by this method should have characteristics from both parent sheep

2. Which of the sheep above might a farmer use to breed sheep with good meat and good wool?
3. Which of the sheep might a farmer use to produce sheep with good wool that can survive in the Welsh mountains?

A sheep farmer has a flock of Wensleydale sheep. To win a prize at the local farming show, he wants to have sheep with very long wool. Some time ago, he took the ewes (female sheep) and rams with the longest wool and allowed them to breed. He allowed the new lambs to grow and then kept the ones with the longest wool and allowed them to breed. The others were sold at market. It took him eight years to produce sheep with wool long enough for him to win a prize.



4. What is a male sheep called?
5. Why is this farmer's method an example of selective breeding?
6. Suggest why it took him so long to win a prize.
7. Think about the weather conditions where you live. Write down a list of characteristics that a sheep should have if it were to be farmed in your area.
8. What other characteristics would you like your sheep to have and why?

A: able to live in cold, harsh conditions. B: produces good quality wool. C: produces good quality meat. D: produces a lot of milk.

The Bluefaced Leicester (good quality wool) and the Polled Dorset (good quality meat).

The Balwen mountain sheep (can live in cold, harsh conditions) and the Bluefaced Leicester (good quality wool).

A ram.

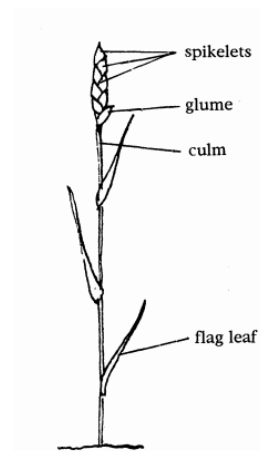
Because he chose the sheep with the longest wool and only bred those sheep together, so the useful characteristic (long wool) was passed on and improved.

The Plant Breeding Game

This game is played in a similar way to 'rummy'. 2, 3 or 4 people can play.

1 Shuffle the pack of cards.

2 Deal cards face down until each player has 5 cards.



The 'hand' of five cards you hold represent a crop of wheat. The cards tell you the characteristics of your crop. For example, one card might show that your crop has few spikelets per ear - a poor characteristic, because it means fewer seeds from each plant, and so a lower value crop. The object of the game is to exchange cards in order to improve your crop.

3. Deal one card face-up in the centre of the table and place the remaining pack of cards face down beside it.

4. The player on the left of the dealer starts.

5 When it is your turn, you may do one of three things: • pass (Le.do nothing!) • exchange one of your cards for the top card on the 'face up pile' • put one of your cards onto the 'face-up' pile and take the top card from the 'face-down' pile The turn then passes to the next player on your left.

6. After the turn has gone round six: times, the game is over. Each player must lay down their hand face up.

7 Use the scoring sheet to find out how much your crop is worth. The player with the most valuable crop is the winner!

Plant Breeding Game Score Sheet

Each of the 5 cards in your 'hand' contributes something to the value of your crop. Use the table below to look up the value of each of your cards. Add the values together to find the value of seed from your new wheat variety in HK\$ per tonne. (But, if you have two or more cards for anyone attribute - e.g. 2 cards for brittleness - then you can only count the average value of these, not the total)

Crop attribute	Value	Crop attribute	Value
FEW spikelets per ear	HK\$ 0	VERY HAIRYglumes	HK\$ 5
AVERAGE spikelets per ear	HK\$ 15	MEDIUM glumes	HK\$ 5
MANY spikelets per ear	HK\$ 30	NON HAIRYglumes	HK\$ 5
LIGHT grain	HK\$ 5	PALE coloured glumes	HK\$ 5
AVERAGE grain	HK\$ 15	MEDIUM coloured glumes	HK\$ 5
HEAVY grain	HK\$ 35	DARK coloured glumes	HK\$ 5
EASY to thresh	HK\$ 20	VERY STIFF culms	HK\$ 25
MEDIUM for threshing	HK\$ 10	MEDIUM culms	HK\$ 15
HARD to thresh	HK\$ 5	WEAK culms	HK\$ 5
LARGE flag leaf	HK\$ 25	GOOD germination rate	HK\$ 15
MEDIUM flag leaf	HK\$ 15	MEDIUM germination rate	HK\$ 5
SMALL flag leaf	HK\$ 5	POOR germination rate	HK\$ 0
VERY brittle ear	HK\$ 0	GOOD response to fertilizer	HK\$ 30
MEDIUM brittle ear	HK\$ 10	AVERAGE response	HK\$ 15
NON brittle ear	HK\$ 25	POOR response	HK\$ 5
EYESPOT resistant	HK\$ 10	GOOD for bread making	HK\$ 20
MILDEW resistant	HK\$ 20	AVERAGE for bread making	HK\$ 10
ERGOT resistant	HK\$ 25	POOR for bread making	HK\$ 0
NOT disease resistant	HK\$ 0	HIGH protein content	HK\$ 20
HIGH shattering	HK\$ 0	MEDIUM protein content	HK\$ 10
MEDIUM shattering	HK\$ 10	LOW protein content	HK\$ 0
LOW shattering	HK\$ 15	EARLY maturation	HK\$ 15
LATE maturation	HK\$ 15	MEDIUM maturation	HK\$ 5

Q1. Many different types of animals are produced using selective breeding.

Some cats are selectively bred so that they do not cause allergies in people.

(a) Suggest **two other** reasons why people might selectively breed cats.

1 .To produce cats with a particular appearance (e.g. fur length/colour, eye colour) for showing.

.....

2 .To produce cats with a friendly temperament that make good pets.....

.....

(2)

(b) Selective breeding could cause problems of inbreeding in cats.

Describe **one** problem inbreeding causes.

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(1)

Inbreeding reduces the gene pool, so harmful recessive alleles are more likely to be passed on and the kittens may have inherited diseases or weak immune systems.

(c) Many people have breathing problems because they are allergic to cats.

The allergy is caused by a chemical called Fel D1.

Different cats produce different amounts of Fel D1.

A cat has been bred so that it does not produce Fel D1.

The cat does **not** cause an allergic reaction.

Explain how the cat has been produced using selective breeding.

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(4)

(Total 7 marks)

Breed the cat that produces no Fel D1 with another cat that produces little or no Fel D1, then keep breeding only the offspring that produce no Fel D1, repeating this over many generations.

BINGO



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Lesson 3 - Natural Selection

- Describe the theory of natural selection.
- Model Natural Selection.

Who is responsible for the theory of evolution by Natural Selection?

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.....
Charles Darwin.

What did he notice about the animals and plants which he was looking at?

Write down the major points from the video, the summary contains all the information.

<https://www.youtube.com/watch?v=UKETe78t5p8>
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Variation

1. Which of these features is an example of continuous variation?

Gender

Blood Group

Weight

Darwin noticed that organisms produce more offspring than can survive, that individuals of a species vary, and that those with useful variations are more likely to survive and reproduce and pass the variation on.

2. Which of these features is an example of discontinuous variation?

Forearm length

Blood group

Height

3. Which of these features shows inherited variation?

Brown eyes

Scars

Pierced ears

4. Which of these features shows environmental variation?

Scars

Blood Group

Eye colour

5. Where is most variation found?

Between species

Within one species

Between lions and tigers

Natural Selection

You will be given a card showing a bird and its matching beak type:



Throughout each round try to 'eat' as much food as you can. You can only use your beak to get beans into your petri dish (stomach).

Butter Bean	1 point	
Black Bean	2 points	
Mung Bean	5 points	

Points to survive:	25
Points to survive and reproduce:	50

Instructions:

- Season 1: You have 30 seconds to 'eat' as much food as you can.
- Season 2: There's a famine, now only half of the food is available. You must compete to get enough food to survive and reproduce! You have 30 seconds to 'eat' as much food as you can.
- Season 3: There's been a change to the environment. Everyone must now use their non-dominant hand. You have 30 seconds to 'eat' as much food as you can.
- Season 4: There's a mutation! Bird number 4 now has a spoon shaped beak. You have 30 seconds to 'eat' as much food as you can.

				
Season 1 – normal breeding				
Season 2 - famine				
Season 3 – change to the environment				
Season 4 - mutation				

Follow-Up Questions:

1. During the ‘normal’ breeding season, was there one bird who was better adapted to collecting food than the other birds? How do you think this would affect the number of birds with this beak in the population?

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Yes - the bird whose beak best matched the food (e.g. tweezers for beans) collected the most. It would survive and reproduce more, so the number of birds with that beak would increase in the population.

2. Describe what happened during the ‘famine’ season. Did any bird outperform the others?

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.....

With only half the food available the birds competed more; the best-adapted beaks still gathered the most food while poorly adapted birds got little, so those birds survived less and produced fewer offspring.

3. How did the change to the environment impact feeding during the third season? How do you think this would affect the number of birds with each beak in the population?

.....

.....

Using the non-dominant hand made feeding harder for every bird, so fewer beans were collected; the birds with the better (more useful) beaks would survive more, so those beaks would become more common.

4. Did the mutation impact feeding during the fourth season? How do you think this would affect the number of birds with each beak in the population?

.....

.....

Yes - the spoon-shaped beak scooped up lots of food, so bird 4 got much more. It would survive and reproduce more, so the spoon-beak mutation would spread and become more common in the population.

Lesson 4 - DNA

- Describe the structure of DNA, using the keyword list below
- Describe the structure of DNA using diagrams.
- Explain how the bases on the two strands link together.
- Describe in simple terms how DNA and genes link to proteins and characteristics

DNA Structure

Fill in the blanks, and use the notes on the next page to help.

DNA has a structure like a twisted ladder. The structure is called a double helix.

DNA has two strands. Each strand is made of units called nucleotides joined together.

Each unit contains three parts: a sugar (deoxyribose), a phosphate and a base.

There are four different bases each represented by the letters A, T, C and G.

The two strands are complementary with A linked T and C linked to G.

As there are many units, the whole structure is called a polymer.

A sequence of three bases is the genetic code for a particular amino acid.

The order of bases controls the order in which amino acids are assembled to produce a particular protein.

When the amino acid chain is complete it folds up to form a protein.

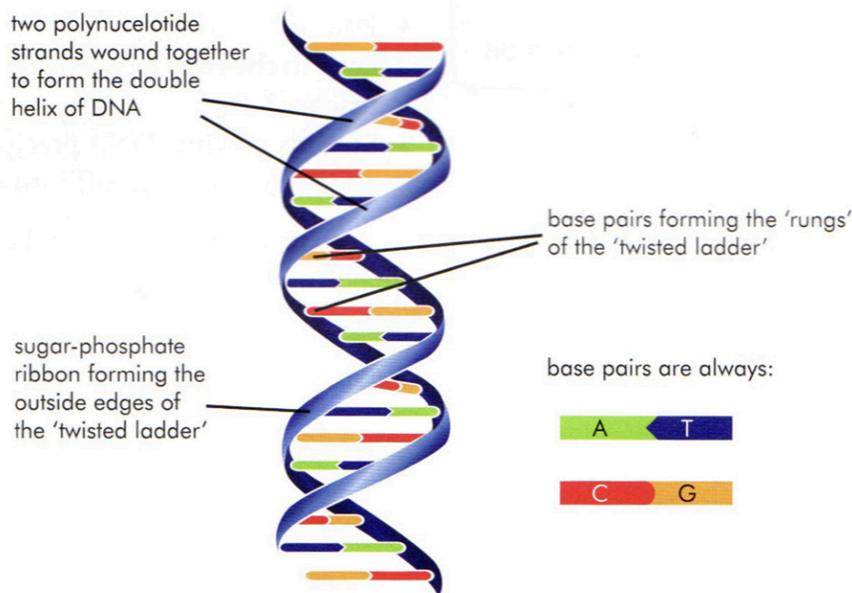
Mutations occur continuously.

A few mutations code for an altered protein with a different shape.

protein, bases, four, amino acids, C, unique shape, two strands, mutations, protein, sugar molecule, DNA, mutations, GACT, A, protein, phosphate group, shape, double helix, T, polymer, G, amino acid, nucleotides.

■ The structure of DNA

A DNA molecule is made of two strands of **polynucleotides** spiralling around one another to form a double helix shape. A nucleotide is made up of a sugar, phosphate and a base. If you imagine the DNA double helix as a twisted ladder, the sides of the ladder consist of an alternating series of sugars (deoxyribose) and phosphates. The rungs of the ladder are composed of pairs of bases. One base of each pair forms half a rung (Figure 5.28).



Each base projects from one side of the ladder, towards the middle of the double helix. Here it matches up with a complementary base, projecting from the opposite side of the ladder. There are four bases, usually known by their initial letters, as follows: adenine (A), thymine (T), guanine (G) and cytosine (C).

The bases can only pair up according to the following rules:

- A can only form a pair with a T
- G can only form a pair with a C.

The double helix structure of DNA was proposed in Cambridge by James Watson and Francis Crick in 1953. Later they were awarded a Nobel Prize for this work. They had solved the riddle of the structure of this important molecule and they also realised how it could be accurately copied (replicated) as part of cell division. This discovery opened the way for enormous advances in understanding how DNA works and in the development of techniques of gene technology that have become an important part of scientific work in the 21st century, with applications in many fields.

When a DNA double helix is split lengthwise into two halves, the missing half of each ladder is rebuilt. Because A always pairs with T, and G always pairs with C, the sequence of bases in the two new complete molecules of DNA is identical. This is crucial when a cell divides by mitosis as it ensures that each new daughter cell gets a full set of chromosomes (and therefore, the same genes) from the original parent cell.

Extracting DNA from Fruit

Introduction

Have you ever wondered how scientists extract DNA from an organism?

All living organisms have DNA, which is short for deoxyribonucleic acid; it is basically the blueprint for everything that happens inside an organism's cells. Overall, DNA tells an organism how to develop and function, and is so important that this complex compound is found in virtually every one of its cells.

In this activity you'll make your own DNA extraction kit from household chemicals and use it to separate DNA from fruit (e.g. strawberries or kiwis).

Background

Whether you're a human, rat, tomato or bacterium, each of your cells will have DNA inside of it (with some rare exceptions, such as mature red blood cells in humans). Each cell has an entire copy of the same set of instructions, and this set is called the genome. Scientists study DNA for many reasons: They can figure out how the instructions stored in DNA help your body to function properly.

We can use DNA to make new medicines or genetically modify crops to be resistant to insects. They can solve who is a suspect of a crime, and can even use ancient DNA to reconstruct evolutionary histories!

To get the DNA from a cell, scientists typically rely on one of many DNA extraction kits available from biotechnology companies. During a DNA extraction, a detergent will cause the cell to pop open, or lyse, so that the DNA is released into solution. Then ethanol added to the solution causes the DNA to precipitate out. In this activity, strawberries or kiwis will be used because each fruit cell has eight copies of the genome, giving them a lot of DNA per cell. (Most organisms only have one genome copy per cell.)

Materials:

Goggles
Fruit (either strawberries or kiwi)
Ice cold ethanol
Water
Spatula
Beaker x2
Table salt (sodium chloride)

Detergent (dishwashing liquid)
Muslin (cheesecloth)
Funnel
Conical flask
Scissors
Resealable plastic sandwich bag
Glass rod

Method:

- 1) Make the 'extraction liquid' by mixing approximately 50cm³ of water with one spatula of salt and a pipette full of detergent in a beaker.
- 2) Either: - collect two strawberries and pull off the green leaves/stalk
Or: - collect half a kiwi and cut off any skin
- 3) Place the fruit into a resealable bag. Push out all of the air and seal the bag.
- 4) With your fingers, squeeze the fruit reducing it to a liquid mush with no lumps.
- 5) Add the extraction liquid, push out all of the extra air and reseal the bag.
- 6) Again, with your fingers, squeeze and mix the fruit mixture for about three minutes.



- 7) Set up the muslin cloth in a funnel ready to filter.
- 8) Cut a small hole in the bottom of the resealable bag and squeeze the mixture from the bag into the funnel. Let it drip through the cloth and into a conical flask.
- 9) Once all the liquid has been filtered, pour the filtrate into a clean beaker.
- 10) Tilt the beaker and very slowly pour approximately 50 cm³ of ethanol down the side so that it 'floats' on the top of the filtrate - try not to allow the fruit mixture and ethanol mix.
- 11) Study the mixture, especially where the two layers meet. The fruit DNA will precipitate out into the ethanol layer and you should see a gooey clear/white stringy material.
- 12) Use a glass rod to pull the DNA out of the mixture and place it onto some paper towel.

Observations and analysis:

1) How does the detergent (in the extraction mixture) affect the fruit cells?

The detergent dissolves the fats (lipids) in the cell membranes, so the cells break open (lyse) and the DNA is released into the liquid.

2) How does the salt affect the DNA in the fruit cells?

The salt cancels out (neutralises) the negative charge on the DNA, so the DNA molecules clump together and fall out of solution (precipitate) more easily.

3) Why does the ethanol need to be poured very slowly into the final beaker?

So the ethanol stays as a separate layer floating on the filtrate - if they mix, the DNA will not precipitate out where the two layers meet.

4) Were you able to see DNA when you added the cold ethanol? Which layer was the DNA in? Describe what it looks like.

Yes - the DNA collected in the ethanol layer (the top layer), as a white/clear, stringy, gooey (slimy) material where the two layers met.

5) DNA is an extremely thin molecule. How is it possible that you can see it?

Although one DNA molecule is too thin to see, billions of DNA molecules clump together (precipitate) in the ethanol, forming a mass big enough to see and to wind onto the glass rod.

Extension: We use strawberries and kiwi fruit as the cells are “octoploid”. What does octoploid mean and how does this compare to human cells? Why do you think strawberries were used in this experiment?

Octoploid means the cells have eight copies (8 sets) of the genome, whereas human cells are diploid (2 copies). So strawberries/kiwis contain much more DNA per cell, making it easier to see.

RECREATE A FACE

DNA key



Gene	DNA clue	Face feature
Eye colour	CTCCTC	brown eyes
	TGTCTC	green eyes
	TGTTGT	blue eyes
Hair colour	GAATGT	blonde hair
	GAAAGG	brown hair
	GATAGG	black hair
Hair type	AGAA TT	curly hair
	AGAT TT	wavy hair
	AGTT TT	straight hair
Freckles	TGGCAT	lots of freckles
	TGGGAT	some freckles
	CGTGAT	no freckles

RECREATE A FACE

Worksheet



Skeleton number: _____

Appearance:

Eye colour	☐ brown	☐ green	☐ blue
Hair colour	☐ blonde	☐ brown	☐ black
Hair type	☐ curly	☐ wavy	☐ straight
Freckles	☐ lots	☐ some	☐ none

· Pupil records their own skeleton: write the skeleton number and name, then tick one option for eye colour, hair colour, hair type and freckles to match that skeleton's DNA clues (see the key on the previous page).

Skeleton name: _____

Lesson 6 - Genes and Mutations

- Define mutation.
- Describe how mutations are the source of genetic variation.
- State the causes of mutations.

Task 1

1. (a) What is the definition of a mutation?

~~A mutation is a random change in the base sequence of DNA.~~.....

(b) Mutations have an effect on the organism by changing the DNA sequence. What is the purpose of DNA in the first place, i.e. what does DNA code for?

~~Proteins - the base sequence codes for the amino acids that make a protein.~~.....

Remember DNA is a code, so if the code is changed, this can cause problems. Or sometimes a benefit, but the characteristics of an organism (phenotype) can change.

Imagine this sentence is DNA

the cat saw the bat

If a letter is swapped the sentence may still make sense.

the cat saw the mat

But DNA is actually 'read' 3 bases at a time.

So what would happen if a base was added or deleted?

the rca tsa wth eba

Adding the r, in completely messed up the sentence.

Same if a letter was deleted.

the cat awt heb at

Deleting the s changed the DNA completely.

2. Look at the following DNA base sequences. What type of mutation has occurred?

Original DNA sequence:CCTAGGAACTGATAC.....

(a) ...CCTAGCAACTGATAC... =

(b) ...CCTAGGACTGATAC... =

(c) ...CGCTAGGAACTGATAC... =

Substitution (base G has been swapped for C)

Deletion (the base A has been deleted)

Insertion (an extra base C has been added)

3. Determine whether the following mutations are beneficial, harmful or have no effect on the survival of the organism.

(a) A seagull with a longer wingspan living on a deserted island.

Beneficial - a longer wingspan gives more lift, so it can glide further to find food.

(b) Roses with a stronger scent grown in a greenhouse.

(c) An albino tiger living in a zoo.

(d) A plant with narrower leaves in a tropical rainforest.

No effect - the rose is grown and cared for in a greenhouse, so a stronger scent gives no survival advantage.

Extension question:

Suggest the type of environment in which the organisms above will have a survival advantage. How does the mutation help the organism to better adapt to the named environment?

A windy island or coast - the longer wingspan gives more lift, so the seagull can glide further to find food.

A hot, dry desert - narrower leaves lose less water by transpiration, so the plant survives drought.

A greenhouse or nursery - humans breed the strongest-smelling roses, so the mutation is selected for.

No effect - the tiger is fed and protected in the zoo, so its white coat does not affect its survival.

Task 2

1. Link the following cancers to their most likely cause:

Harmful - narrower leaves have less surface area, so less light is absorbed for photosynthesis.

Lung cancer	☐	☐	UV radiation
Cervical cancer	☐	☐	alcohol
Skin cancer	☐	☐	human papilloma virus
Liver cancer	☐	☐	poor diet
Colon cancer	☐	☐	air pollution

Air pollution

Human papilloma virus

2. The cancers listed above are known as preventable cancers.

(a) What is a "preventable" cancer?

A cancer caused by lifestyle or environmental factors, so the risk can be lowered by changing behaviour.

(b) Suggest ways by which each of these cancers can be prevented.

Lung - don't smoke; cervical - HPV vaccine and screening; skin - sunscreen and cover up; liver - limit alcohol; colon - healthy diet

(c) Explain why skin cancer is not inherited.

Skin cancer is caused by UV radiation damaging DNA in body (skin) cells during life, so the mutation is not passed on in gametes

Extension question:

Suggest why the risk of developing cancers is linked to age.

~~Mutations build up as cells divide repeatedly over the years, and the body's DNA repair and immune defences become less efficient with age.~~

.....

.....

.....

.....

.....

Plenary:

What are the three key learning points from the lesson?

1.

2.

3.

A mutation is a random change in the base sequence of DNA.

Mutations are the source of genetic variation because they produce new alleles.

Causes include radiation (UV, X-rays) and chemicals such as tar in tobacco; effects can be beneficial, harmful or have no effect.

Lesson 7 - Genetic Crosses

Define the terms genotype, phenotype, dominant and recessive allele, homozygous and heterozygous.

Predict probabilities of outcomes from genetic crosses.

What is a gene?

A gene is a section of DNA that codes for a particular characteristic (protein).

What is an allele?

An allele is a different version of a gene.

Humans inherit **two** copies of every gene you have. One from each parent. So when a characteristic is expressed in you, which trait do you show?

Gregor Mendel discovered that genes can be passed on, but sometimes one is not expressed while the other is, but the not expressed trait can still be passed on again.

What is the definition of.....

Dominant allele

Dominant allele - the allele that is always expressed, even when only one copy is present.

Recessive allele

Recessive allele - only expressed when two copies are present (no dominant allele).

Watch this [youtube video](#) first.

Question 1:

Trait: Brown eyes are dominant, blue eyes are recessive.

Alleles for eye colour can be shown using letters: B = brown eyes, b = blue eyes

Mum only has alleles for brown eyes and Dad only has alleles for blue eyes.

a) Write down the alleles of the parents using the letters for the different eye colours? (Remember, all genes come in pairs)

Mum =

Dad =

b) What colour eyes will mum have?

c) What colour eyes will dad have?

Children inherit half their genes from mum and half from dad. This means that although both mum and dad have two copies of the genes for eye colour, only one copy of the gene from each parent will be inherited by their children. We can show how this happens using a Punnett square.

Separate each parents' genes for eye colour and add dad's in the first row and mum's in the first column:

Mum \ Dad		
B	bb	
Brown eyes - she has two dominant B alleles. Blue eyes - he has two recessive b alleles.		
B		

Now combine the letter in each row with the letter in the corresponding column in the squares. This gives us the potential combinations of genes that the children could inherit:

Mum \ Dad	b	b
B	Bb	Bb
B	Bb	Bb

d) What colour eyes will their children have?

Now try the following exercises....

All the children will have brown eyes, because they all inherit Bb.

Key words:

(a) What does the term **heterozygous** mean?

~~Heterozygous means having two different alleles for a gene (e.g. Bb).~~
.....

(b) What does **homozygous** mean?

~~Homozygous means having two identical alleles for a gene (e.g. BB or bb).~~
.....

(c) What does **genotype** mean?

~~Genotype is the combination of alleles an organism has (e.g. Bb).~~
.....

(d) What does **phenotype** mean?

~~Phenotype is the physical characteristic shown, e.g. brown eyes.~~
.....

Question 2:

(a) Both parents are **heterozygous** for brown eyes.

(i) What is the **genotype** for each parent?

Mum =

Dad =

(ii) What will be the genotypes of their children? Fill in the Punnett square below.

Bb

Bb

Mum \ Dad	B	b
B	BB	Bb
b	Bb	bb

(c) What is the probability that their children will have brown eyes?

75% (3 out of 4 children)

Extension Question:

In pea plants, height is also controlled by a gene where tall is dominant and short is recessive.

(a) Select a letter for height and write down the key for the alleles:

T = tall (dominant), t = short (recessive)

(b) A short plant was crossed with a heterozygous tall plant. What are the genotypes for the parents?

Short parent =

Tall parent =

(c) Complete the Punnett square below for the inheritance cross.

		tt	
Tt	Tall \ Short	t	t
	T	Tt	Tt
	t	tt	tt

(d) How many different genotypes are possible in the offspring?

2 (Tt and tt)

(e) What phenotypes are possible in the offspring?

Tall and short plants (50% tall, 50% short)

(f) What is the probability for the offspring being short?

50% (1 in 2)

1. A heterozygous rabbit is crossed with a homozygous dominant rabbit. What are the results? (B = grey b = white).

Bb x BB: 50% BB and 50% Bb, so all the offspring are grey.

2. In cats having hair is dominant (H) over not having hair (h). If a male cat that is heterozygous for hair is crossed with a hairless cat what are the genotypes and phenotypes that their offspring may have?

Hh x hh: 50% Hh (with hair) and 50% hh (hairless).

3. A homozygous dominant tomato plant is crossed with a homozygous recessive tomato plant. What are the results? (T = tall t = short)

TT x tt: all the offspring are Tt, so 100% are tall.

4. If two black mice are crossed, ten black and three white mice result. a. Which allele is dominant? b. Which allele is recessive? c. What are the genotypes of the parents?

a. Black is dominant. b. White is recessive. c. Both parents are heterozygous (Bb x Bb).

5. If long floppy ears is dominant (L), over short spiked ears (l) and you find a puppy with long floppy ears, how could you determine the puppy's genotype?

Do a test cross with a short-eared (ll) dog: if all the pups have long floppy ears it is LL; if about half are short-eared it is Ll.

6. Tongue rolling is a dominant trait. Can a husband and wife who are not able to roll their tongues produce a child that is able to roll his or her tongue?

No - tongue rolling is dominant, so non-rollers must be homozygous recessive (tt x tt) and all their children would be tt (non-rollers).



Name:

Date:

Student Exploration: Mouse Genetics (One Trait)

Directions: Follow the instructions to go through the simulation. Respond to the questions and prompts in the orange boxes.

Vocabulary: allele, DNA, dominant allele, gene, genotype, heredity, heterozygous, homozygous, hybrid, inheritance, phenotype, Punnett square, recessive allele, trait

Prior Knowledge Questions (Do these BEFORE using the Gizmo.)

1. The image shows a single litter of kittens. How are they similar to one another?

2. How do they differ from one another?

~~They are alike because they are from the same litter – they have similar fur, eyes and body shape.~~

They differ in fur colour and pattern (black, white or patches).



3. What do you think their parents looked like?

~~One parent was probably black and the other white, or both had black fur but carried the white allele.~~

Gizmo Warm-up

Heredity is the passage of genetic information from parents to offspring. The rules of **inheritance** were discovered in the 19th century by Gregor Mendel. With the *Mouse Genetics (One Trait)* Gizmo™, you will study how one **trait**, or feature, is inherited.

1. Drag two black mice into the **Parent 1** and **Parent 2** boxes. Click **Breed** to view the five offspring of these parents.

What do the offspring look like?

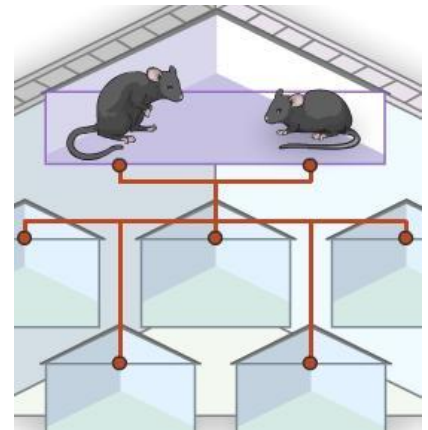
The appearance of each mouse is also called its **phenotype**.

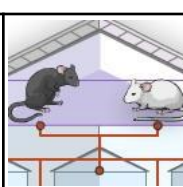
2. Click **Clear**, and drag two white mice into the parent boxes. Click **Breed** several times. What is the phenotype of the offspring now?

~~All the offspring are white.~~
All five offspring are black, like both parents.

3. Do you think mouse offspring will always look like their parents? Explain:

~~No - each parent passes on one allele, so a colour that is hidden in the parents can reappear in their offspring.~~



Activity A: Patterns of inheritance	<u>Get the Gizmo ready:</u> - Click Clear. - Drag a black mouse and a white mouse into the parent boxes, but don't click Breed yet.	
--	--	---

Question: What patterns are shown by offspring traits?

1. Predict: What do you think the offspring of a black mouse and a white mouse will look like?

All the offspring will be black, because black fur is dominant.

2. Observe: Click **Breed** several times. What do you see?

All the offspring are black (they are hybrids, Ff).

3. Observe: Drag two offspring into the **Holding Cages**. These mice are called **hybrids** because their parents had different traits. Click **Clear**, and then breed the two hybrids.

What do you see now?

4. Experiment: Turn on **Show statistics**. Click **Breed** until there are 100 offspring.

Now there is a mixture: about 3 black offspring for every 1 white.

How many offspring were black?

How many were white?

5. Explore: Try other combinations of mouse parents. Write the results of each experiment in your notes. When you have finished, answer the following questions. (Note: You can refer to the parents as "pure black," "pure white," or "hybrid.")

A. Which parent combination(s) yield only white offspring?

B. Which parent combination(s) yield only black offspring?

C. Which parent combination(s) yield a mixture of black and white offspring?

About 75

Hybrid x hybrid and hybrid x pure white (Ff x Ff, Ff x ff)

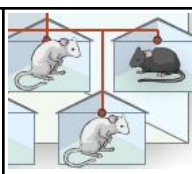
6. Challenge: Based on experiments similar to these, Gregor Mendel devised a theory of inheritance. Use your own observations to come up with your own explanation of how a trait such as fur color is passed down from parents to offspring. If possible, discuss your theory with your classmates and teacher.

About 25

Pure white x pure white (ff x ff)

Pure black x pure black, pure black x pure white or pure black x hybrid (FF x FF, FF x ff, FF x Ff)

Each parent has two alleles for fur colour and passes one at random to each offspring. Black (F) is dominant over white (f), so only ff mice are white.

Activity B: Genetics basics	Get the Gizmo ready: - Click Clear. - Drag a black mouse and a white mouse into the parent boxes.	
--	---	---

Introduction: Inherited traits are encoded on a molecule called **DNA** (deoxyribonucleic acid). **Genes** are segments of DNA that control a particular trait. Most genes have several different versions, or **alleles**. The **genotype** is the allele combination an organism has.

Question: How do alleles determine fur color?

1. **Observe:** Turn on **Show genotype**. Move your cursor over a mouse to see its genotype.

A. What is the genotype of the black parent? White parent?
These mice are **homozygous** for fur color, meaning both alleles are the same.

B. Click **Breed**. What is the genotype of the offspring mice? **Ff**
These mice are **heterozygous** for fur color, meaning the alleles are different.

2. **Analyze:** **Dominant alleles** are always expressed when present. **Recessive alleles** are not expressed when the dominant allele is also present. Look at the two alleles for fur color.

A. Which allele is dominant, and which fur color does it produce?
B. Which allele is recessive, and which fur color does it produce? **f is recessive and produces white fur (only when two f alleles are present).**

FF

3. **Predict:** Place two of the **Ff** offspring into the **Holding Cages**. Click **Clear**, and then place them into the parent boxes.

A. Which allele(s) could the offspring inherit from parent 1?
B. Which allele(s) could the offspring inherit from parent 2?
C. What are the possible genotypes of the offspring? **FF, Ff and ff**
F (the black allele) is dominant and produces black fur.

ff

4. **Experiment:** Click **Breed** several times, and look at the genotypes of the offspring. Did you find all the predicted genotypes? Explain. **Yes - all three genotypes (FF, Ff and ff) appeared, in roughly the ratio 1:2:1.**

F or f

F or f

Activity C: Modeling inheritance	<u>Get the Gizmo ready:</u> - Click Clear. - Drag a black mouse and a white mouse into the parent boxes.	
---	--	---

Question: How do scientists predict the genotypes of offspring?

1. Model: Scientists use a **Punnett square** to model the different possible offspring genotypes from a parent pair. The parent genotypes are written across the top and side of the square, as shown. The four possible offspring genotypes are then filled in.

	F	F
f	Ff	
f		

The first square is filled in for you. Fill in the remaining squares.

- A. What are the genotypes of the offspring?
- B. What percentage of the offspring will have black fur?
- C. What percentage of the offspring will have white fur?

75%
25%

2. Experiment: Click **Breed** several times.

Were your predictions correct?

FF, Ff, Ff and ff

Yes - the results were roughly 75% black and 25% white.

3. Model: Use the Punnett squares below to model each parent combination. After filling in each Punnett square, predict the percentages of black and white offspring.

Parent 1: Heterozygous black (**Ff**)

Parent 2: Heterozygous black (**Ff**)

Predicted % black offspring: _____

Predicted % white offspring: _____

75%

25%

Parent 1: Heterozygous black (**Ff**)

Parent 2: Homozygous white (**ff**)

Predicted % black offspring: _____

Predicted % white offspring: _____

50%

50%

4. Experiment: Turn on **Show statistics** and **Show as approximate percentage**. For each combination, breed approximately 500 offspring. Record the results in the table below.

(Hint: To obtain an *Ff* mouse, breed an *FF* mouse to an *ff* mouse. Place two *Ff* offspring into the holding cages, click **Clear**, and then drag the *Ff* mice into the parent boxes.)

Parent 1 Genotype	Parent 2 Genotype	% Black offspring	% White offspring
<i>Ff</i>	<i>Ff</i>	75%	25%
<i>Ff</i>	<i>ff</i>	50%	50%

5. Draw conclusions: How well did the Punnett squares predict the offspring percentages for each parent pair?
Very well - the predicted 75%/25% and 50%/50% were close to the actual results, although real results vary because fertilisation is random.

6. Summarize: In your own words, describe what heredity is and how it works in mice.

Heredity is the passing of genetic information from parents to offspring. Each mouse has two alleles for fur colour and passes one at random to each offspring; the dominant allele decides the colour.

7. Think and discuss:

Do you think most traits are inherited the way mouse fur color is? Why do you think this is?

No - most traits are controlled by many genes and are also affected by the environment, so they show a continuous range of variation.

Lesson 8 - Genetic Diseases

Describe what is meant by a genetic disorder.

Predict probabilities of outcomes from genetic crosses of recessive or dominant genetic disorders.

- 1) A different version of a gene
- 2) Traits that will always be expressed
- 3) A section of DNA that codes for a particular trait
- 4) Having two alleles that are the same
- 5) The alleles that an individual has

Activity 1: Watch the youtube clip on [Inherited Genetic Disorders](https://www.youtube.com/watch?v=UC-YwiB_FFg)
(https://www.youtube.com/watch?v=UC-YwiB_FFg)



Q1) What is a genetic disorder?

~~A genetic disorder is a disease caused by a faulty gene (a mutation) that is inherited from the parents.~~
Allele

Q2) What type of allele causes polydactyly?

Dominant allele

~~A dominant allele - only one copy is needed for the person to have extra fingers or toes.~~

Gene

Q3) Cystic fibrosis is caused by a recessive allele.

Homozygous

- (a) Two parents are both **carriers** for the disease. What percentage of their children would be likely to inherit cystic fibrosis? Use a Punnett square to show the pattern of inheritance.

Genotype

25% (1 in 4) - only the ff child has cystic fibrosis.

Mum \ Dad	F	f
F	FF	Ff
f	Ff	ff

- (b) What would be the probability of a child inheriting cystic fibrosis if one parent is normal but the other parent is homozygous recessive? Use a diagram to support your answer.

0% - every child would inherit F from the normal parent and f from the homozygous recessive parent, so all would be carriers (Ff).

Mum \ Dad	f	f
F	Ff	Ff
F	Ff	Ff

- Q4) Outline the similarities and differences between polydactyly and cystic fibrosis in how each genetic disorder is inherited.

Similarities	Differences
Both are caused by a mutation in a gene and both are inherited from the parents through the gametes.	
	Polydactyly is caused by a dominant allele, so one copy is enough and it cannot be carried; cystic fibrosis is caused by a recessive allele, so two copies are needed and carriers are healthy.

Activity 2:

- (a) Both parents are **heterozygous** for brown eyes.

- (i) What is the **genotype** for each parent?

Mum =

Dad =

- (ii) What will be the genotypes of their children? Fill in the Punnett square below.

Bb

Bb

Mum \ Dad		b
B		Bb
b		bb

- (c) What is the probability that their children will have brown eyes?

75% (3 out of 4 children will have brown eyes)

Genetic Modification

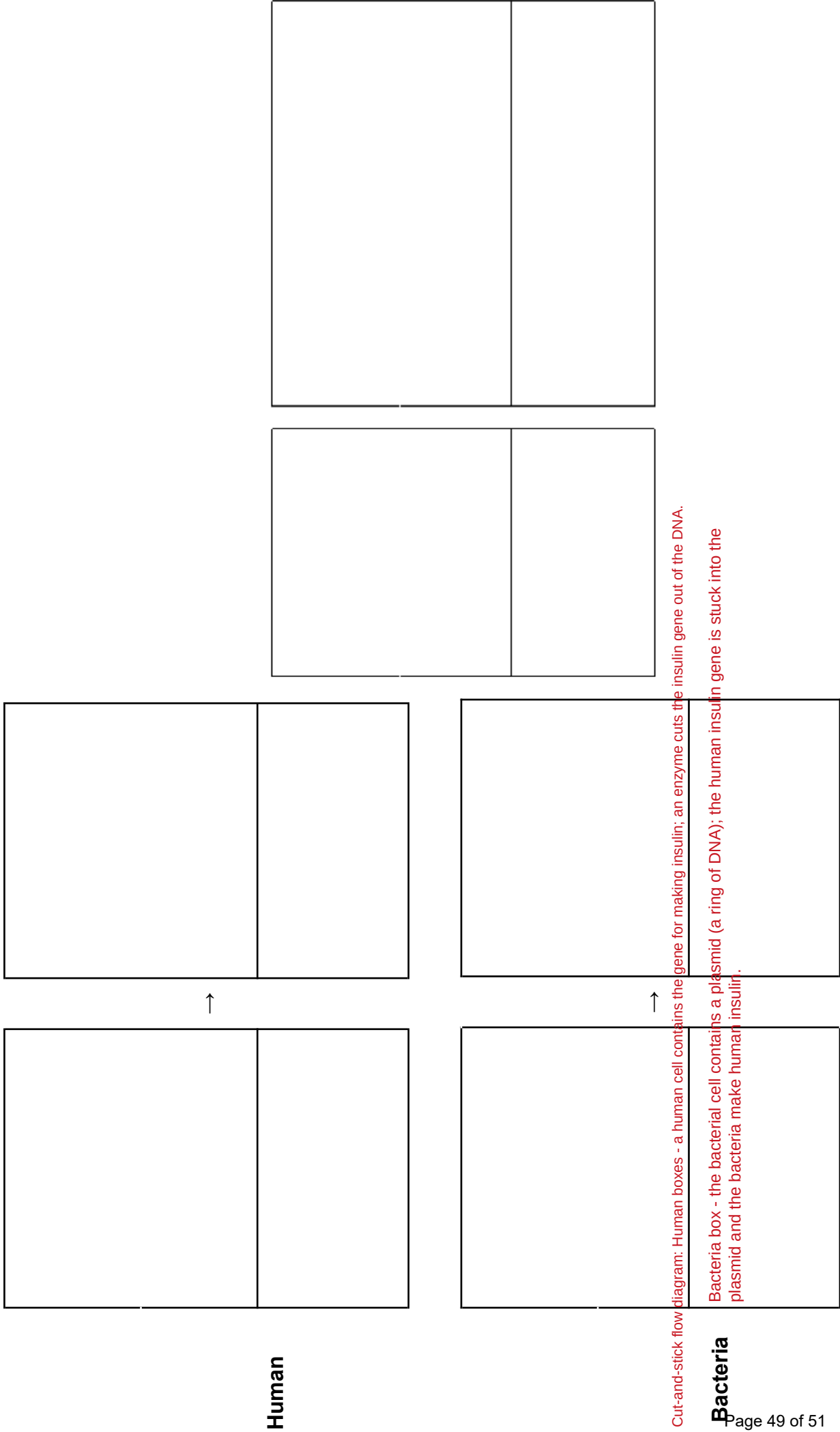
1. Use this space to write some notes about genetic modification:

Genetic modification means changing an organism's DNA, e.g. cutting a useful gene out with an enzyme and sticking it into another organism (a plasmid) with a different enzyme.

2. Cut and stick your work about plasmids on the back of this page.

3. Fill in this table with some benefits and drawbacks of genetically modifying organisms:

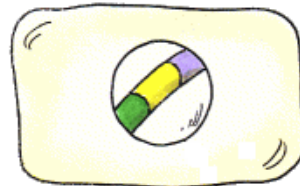
Positives	Negatives
Crops can be made resistant to pests and diseases, so yields increase and less pesticide is needed.	The modified gene could escape into wild plants and affect other organisms and ecosystems.
Crops can be given extra vitamins and nutrients, which improves health and reduces deficiency diseases.	Some people worry about the long-term effects of GM food on human health (unknown risks).
Bacteria can be modified to make human insulin and other medicines cheaply and in large quantities.	GM seeds are expensive for farmers, and some people think it is unethical to change an organism's DNA.



Use a different enzyme to stick the human insulin gene into the bacteria plasmid.



· Cut-and-stick order for the flow chart: (1) Human cells contain a gene for making insulin. (2) Use an enzyme to cut the insulin gene out of the DNA.

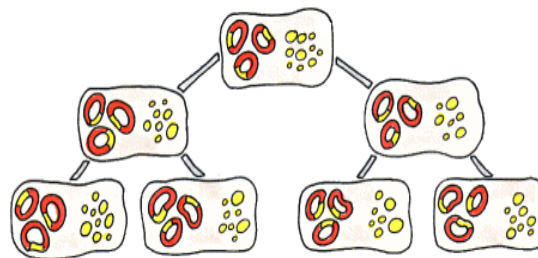


human cell



bacterial cell

The bacteria reproduce and start to make human insulin.



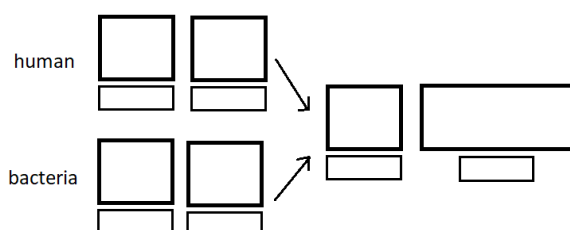
Bacteria cells contain a ring of portable DNA called a plasmid.

Human cells contain a gene for making insulin.

Use an enzyme to cut the insulin gene out of the DNA.

Use the same enzyme to cut bacteria DNA plasmid.

Layout:



· Layout: (3) Bacteria contain a plasmid; the same enzyme cuts the plasmid. (4) A different enzyme sticks the human insulin gene into the plasmid. (5) Bacteria reproduce and make insulin.

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p7: Bar chart: blood group (A, B, AB, O) on the x-axis, percentage of population on the y-axis - bars of 42, 10, 4 and 45.

p8: Line graph: height (cm) on the x-axis, number of people on the y-axis - plot at group mid-points, values 0, 5, 9, 16, 10, 6, 4, 1.

p11: Cross a Golden Retriever (kind, patient, easy to train, good with children) with a long-haired light-coloured dog such as a Rough Collie (65-70 cm, obedient).

p15: Mate the bull that survives temperate climates and produces lots of meat with the cow that survives temperate climates and produces lots of milk.

p16: Wool length only increases by a small amount in each generation, so many generations (about eight years) of selection were needed to get wool long enough to win.

p16: Thick, waterproof fleece to stay warm and dry; hardy; able to find food on rough grazing; good hooves for hilly ground; dark colour for camouflage.

p16: Good quality wool and lots of meat (large, fast-growing) so the flock gives a good yield and is profitable, plus a calm temperament that is easy to handle.

p32: Poor diet